Human–AI Collaboration for UX Research

Designing, Evaluating, and Improving LLM Agents as Simulated Users

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Introduction

About Me

Yuxuan (Leo) Lu https://yuxuan.lu
Ph.D. student, Northeastern University (Khoury) — advised by Prof. Dakuo Wang

Industry Experience: Applied Scientist Intern at Amazon (Sep 2024–Jul 2026); ML Research Intern at LinkedIn China (Jul 2022–May 2023).
Research Focus: I work at the intersection of NLP and HCI.

Selected Works

NLP / ML		**HCI**	
Finetuned LLM Agent	ACL'26	UXAgent	CHI'25 LBW
WebServ	ARR†	+ full system	in submission
Firefly	NeurIPS'26	Agent A/B	CHI'26 Poster

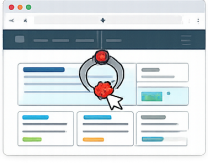
24 selected papers, 6 first-author (above; †under review).

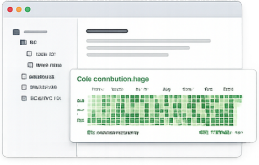
18 co-authored — 15 accepted (ACL'26 $\times$3, ICLR'26, NeurIPS'26, NAACL'24, EMNLP'23/'24/'26, CHI'24/'25, CSCW'24/'26, CoRL'26, ECCV'26) and 3 under review (ARR $\times$2, +1 in submission).

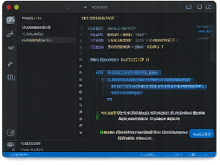
OpenClaw

opencode

Claude Code







Greeting and quick check-in

- Shorten `under review (ACL'26)` to `ACL'26 under review`
- Even shorter: `ACL'26 submission`
- If you want to keep the full phrasing, split it across two lines inside the cell
- If needed, give the right column a bit more width with weighted `X` columns

I'd recommend this version first:

```
\begin{tblr}{
  colspec={X[1.4,l]X[0.9,r]},
  row{1}={font=\bfseries}
}
  NLP & Venue/Status \\
  SFT LLM Agent & ACL'26 under review \\
  WebServ & ACL'26 under review \\
\end{tblr}
```

Why this helps:

- `ACL'26 under review` is shorter and reads more naturally on slides

Ask anything...

Build · GPT-5.4 · Default

Review 3

Session changes · Unified · Split · Collapse all

files/talks/3_papers/talk.tex +8 -1

```
{0.48\linewidth}                    {0.48\linewidth}
26 unmodified lines
204     \centering                  204     \centering
205                                 205
\includegraphics[wid                \includegraphics[widt
th=.8\linewidth]                    h=.8\linewidth]
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206   \end{figure}                  206   \end{figure}
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208 }                               215 }
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    {Today's topic}                     topic}
211   one minute                    218   one minute seperate
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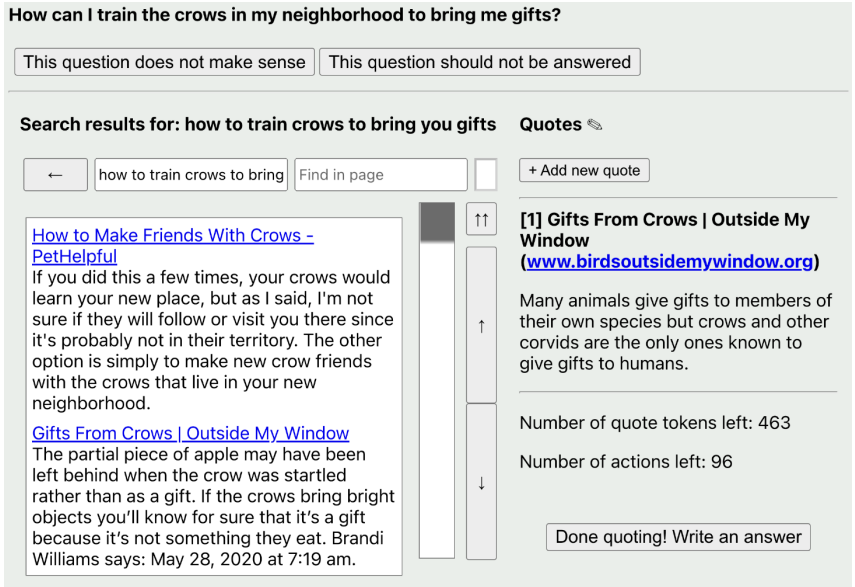
3 Changes · All files

- files
  - talks
    - 3_papers
      - figures
        - opencode.png A
      - talk.pdf M
      - talk.tex M


How can I train the crows in my neighborhood to bring me gifts?
This question does not make sense
This question should not be answered
Search results for: how to train crows to bring you gifts
Quotes
←
how to train crows to bring
Find in page
+ Add new quote
How to Make Friends With Crows - PetHelpful
If you did this a few times, your crows would learn your new place, but as I said, I'm not sure if they will follow or visit you there since it's probably not in their territory. The other option is simply to make new crow friends with the crows that live in your new neighborhood.
Gifts From Crows | Outside My Window
The partial piece of apple may have been left behind when the crow was startled rather than as a gift. If the crows bring bright objects you'll know for sure that it's a gift because it's not something they eat. Brandi Williams says: May 28, 2020 at 7:19 am.
↑↑
↑
↓
[1] Gifts From Crows | Outside My Window (www.birdsoutsidemywindow.org)
Many animals give gifts to members of their own species but crows and other corvids are the only ones known to give gifts to humans.
Number of quote tokens left: 463
Number of actions left: 96
Done quoting! Write an answer

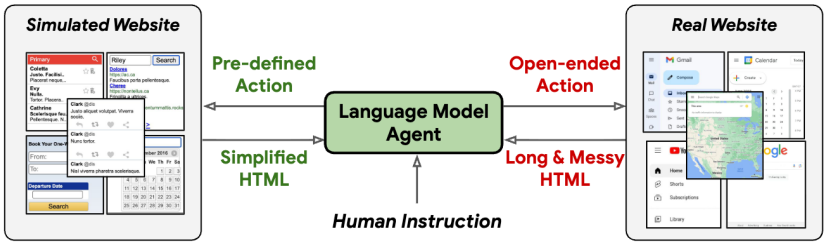
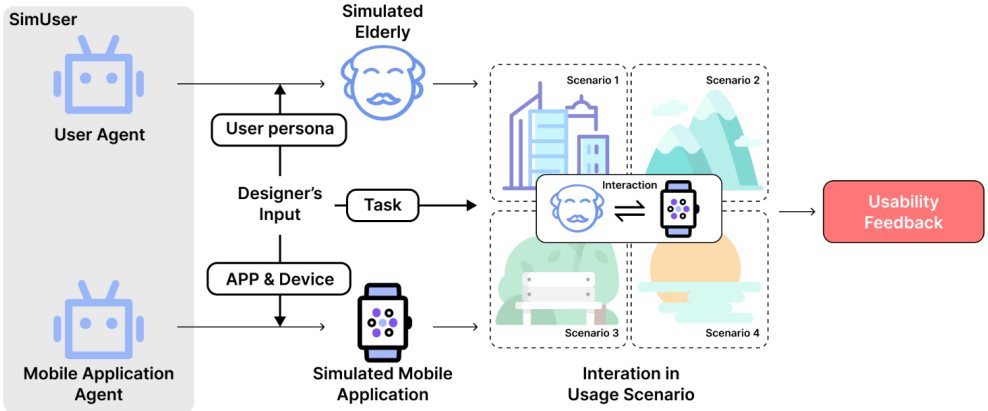


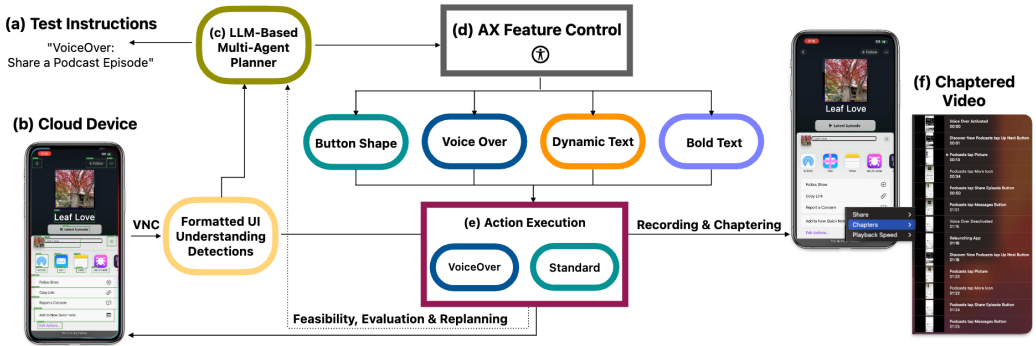
Figure 1: WebAgent, ICLR 2024

Gur et al., 2023, *The Twelfth International Conference on Learning Representations*

Joining for coffee at a cafe
KM:
AC:
[Abigail]: Hey Klaus, mind if I join you for coffee?
[Klaus]: Not at all, Abigail. How are you?
Taking a walk in the park
SM:
Arriving at school
AK:
Sharing news with colleagues
JL:
TK:
[John]: Hey, have you heard anything new about the upcoming mayoral election?
[Tom]: No, not really. Do you know who is running?
Finishing a morning routine
JM:

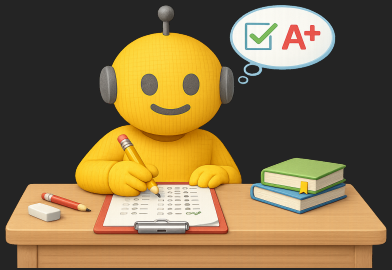


Xiang et al., 2024, *Proceedings of the CHI Conference on Human Factors in Computing Systems*

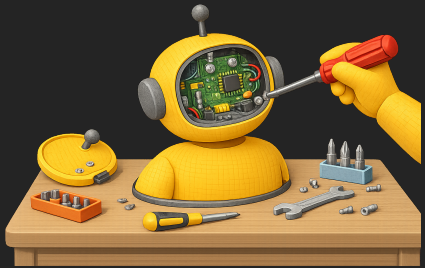

(a) Test Instructions
"VoiceOver:
Share a Podcast Episode"
(c) LLM-Based Multi-Agent Planner
(d) AX Feature Control
(b) Cloud Device
Button Shape
Voice Over
Dynamic Text
Bold Text
VNC
Formatted UI Understanding Detections
(e) Action Execution
VoiceOver
Standard
Recording & Chaptering
(f) Chaptered Video
Feasibility, Evaluation & Replanning

How can we design LLM agent systems for UX research?

How can we evaluate LLM agent systems?



How can we improve LLM agent systems?



Case Study 1: UXAgent: A System for Simulating Usability Testing of Web Design with LLM Agents

What’s the worst nightmare for a researcher?

What's the worst nightmare for a researcher?

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- Spending weeks on a study, only to find that the results are not significant
- Training a model for days, only to find a bug in your preprocessing pipeline
- The myth of "Reviewer 2"

What's the worst nightmare for a researcher?

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- Spending weeks on a study, only to find that the results are not significant
- Training a model for days, only to find a bug in your preprocessing pipeline
- The myth of "Reviewer 2"
- Realizing that your experiment design is flawed one week before deadline

As someone from both NLP and HCI, I think ...

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	NLP	HCI
Experiment Subject	Models and Machines	Human Subjects
Experiment Design	Code and Data	Study Protocol
Experiment Cost	Money	Human Participants' Time
"Debugging" Method	Code Debugging	???

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	NLP	HCI
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"Debugging" Method	Code Debugging	???

Human Participants' Time is Valuable and Limited

Existing LLM Agent systems mostly works in **sandboxed environments**

Challenges

How can we better evaluate UX Research study design before running the study?

How can we better evaluate usability testing study design before running the study

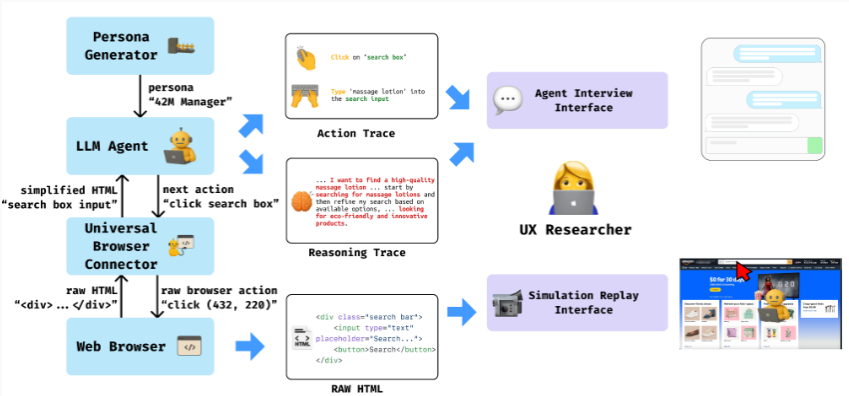
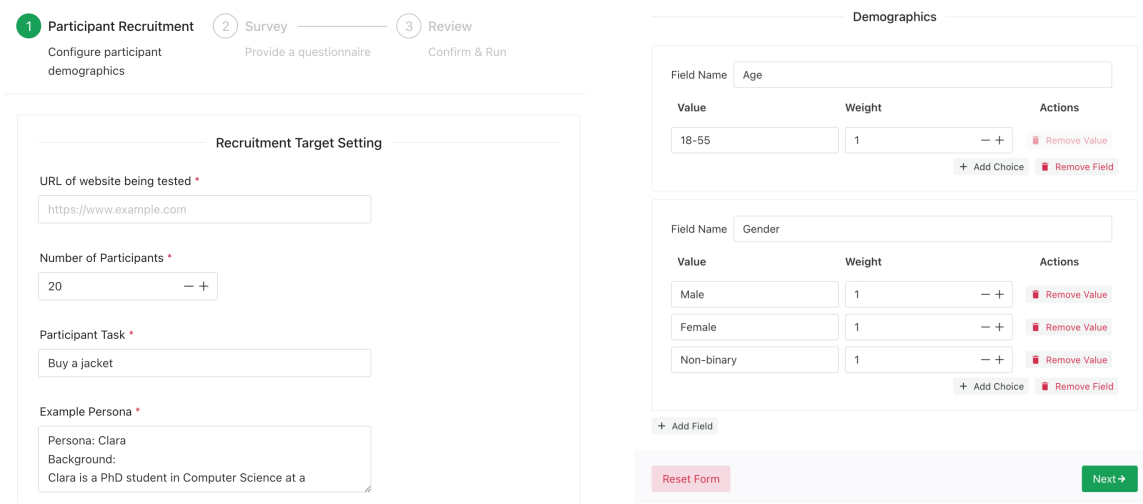


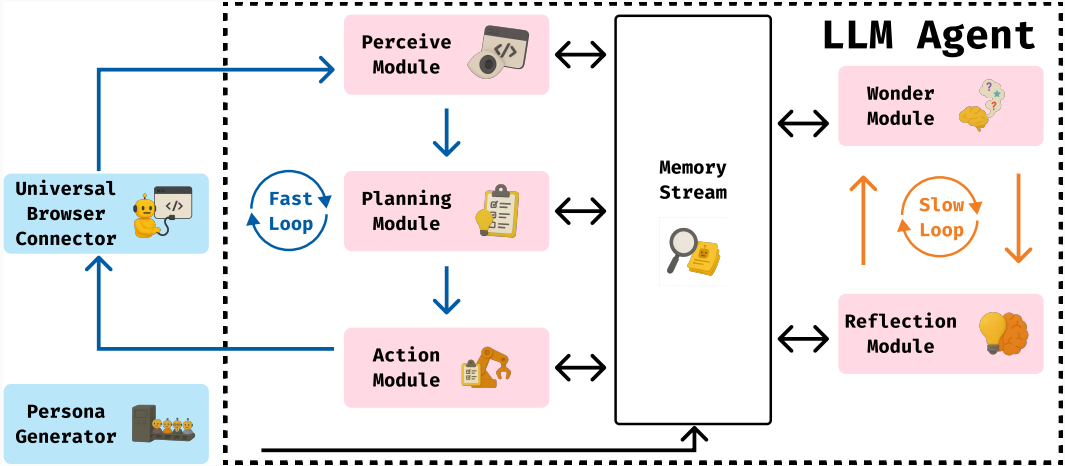
Figure 2: System Architecture of UXAgent

https://www.youtube.com/watch?v=-2xpeJ04mRA

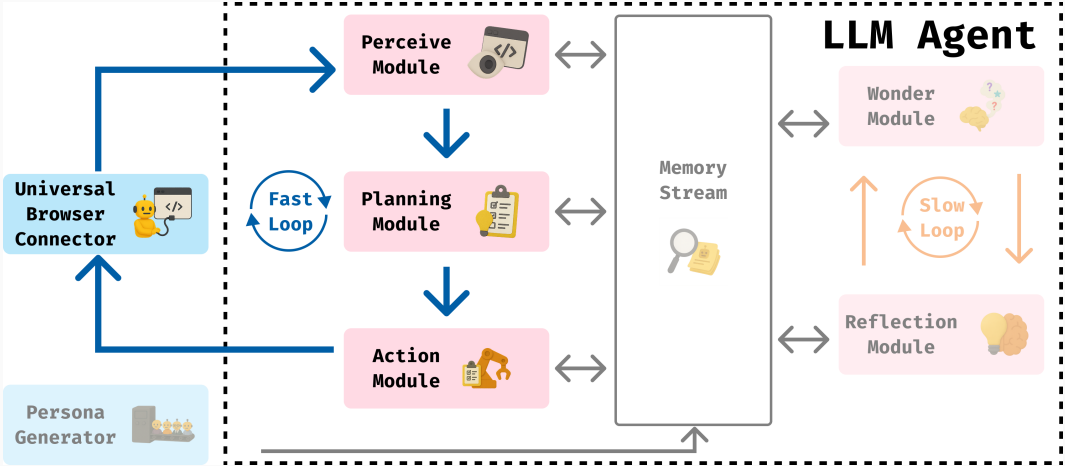


(a) Participant Task Config **(b) Demographic Distribution Config**

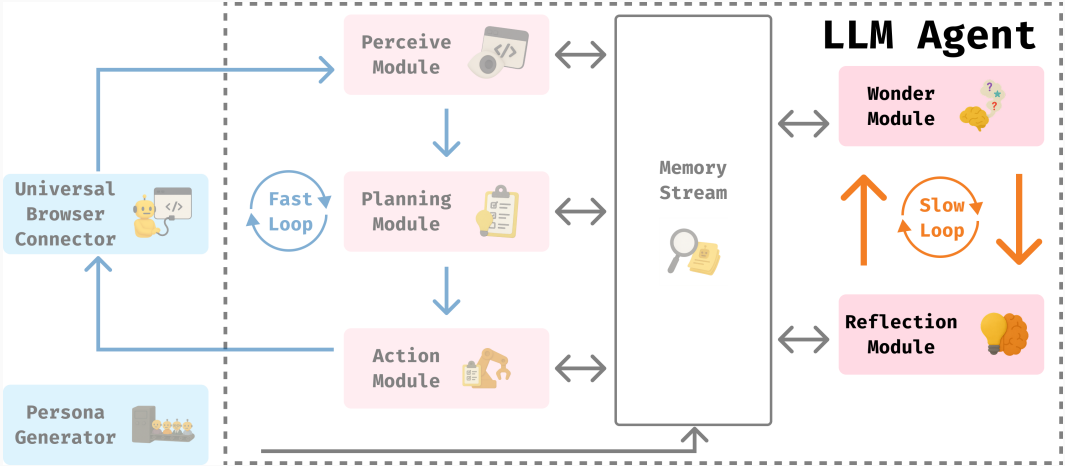
Agent Architecture



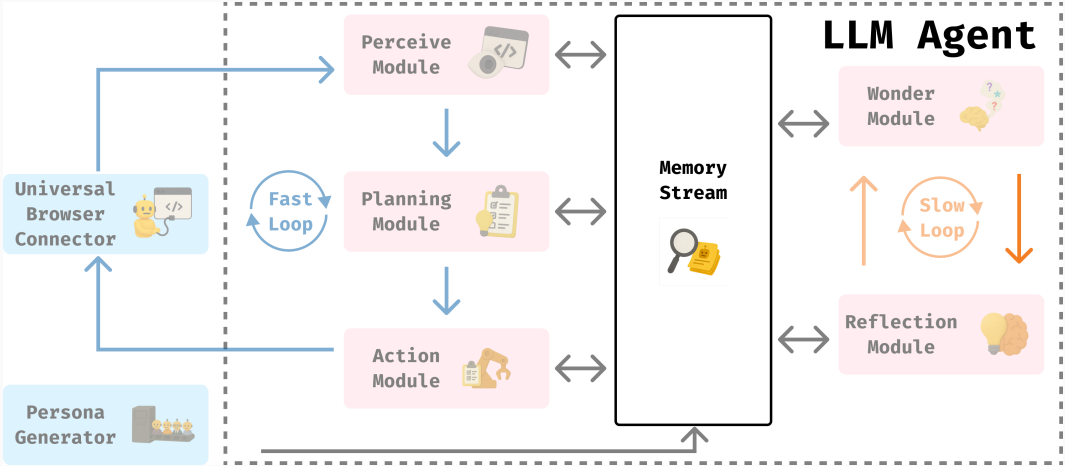
Agent Architecture - Fast Loop



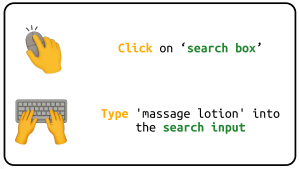
Agent Architecture - Slow Loop



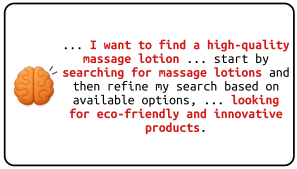
Agent Architecture - Memory Stream



Agent Architecture - Collected Data



Action Trace



Reasoning Trace

RAW HTML

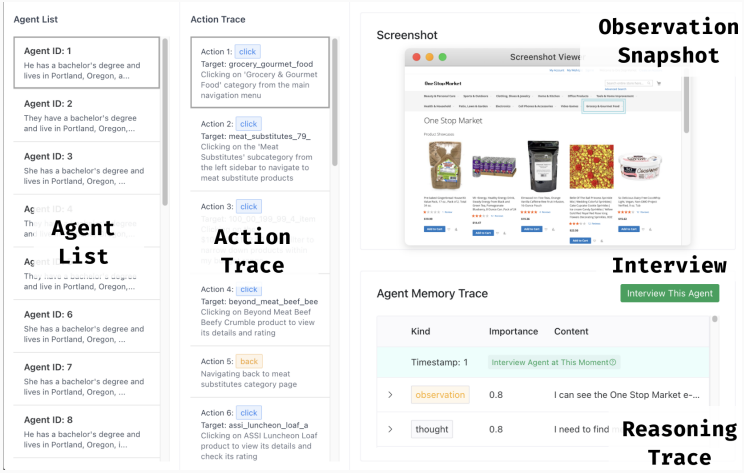


Figure 3: Result Viewer Interface

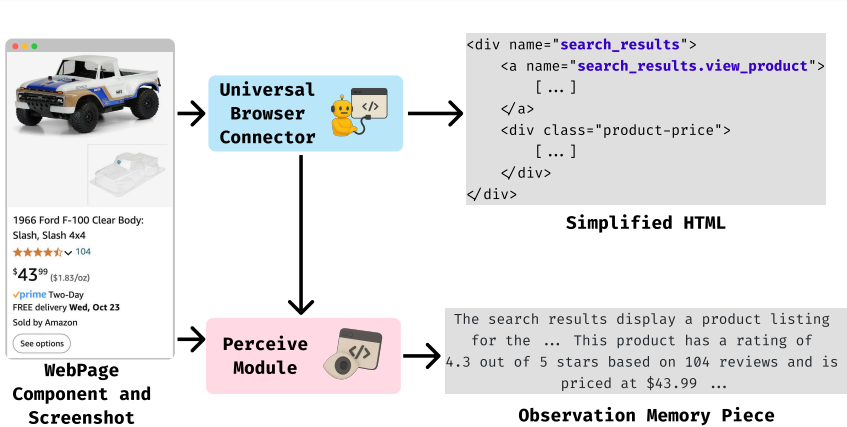


Figure 4: Universal Browser Connector

UXAgent Evaluation: can simulated agents help UX researchers?

User Study Setup

16 UX researchers · 20 simulated sessions · ~30 min / session

Task: buy a meat substitute (top-rated, $100–200), stop at checkout

Post-Study Ratings (Likert -2 to 2)

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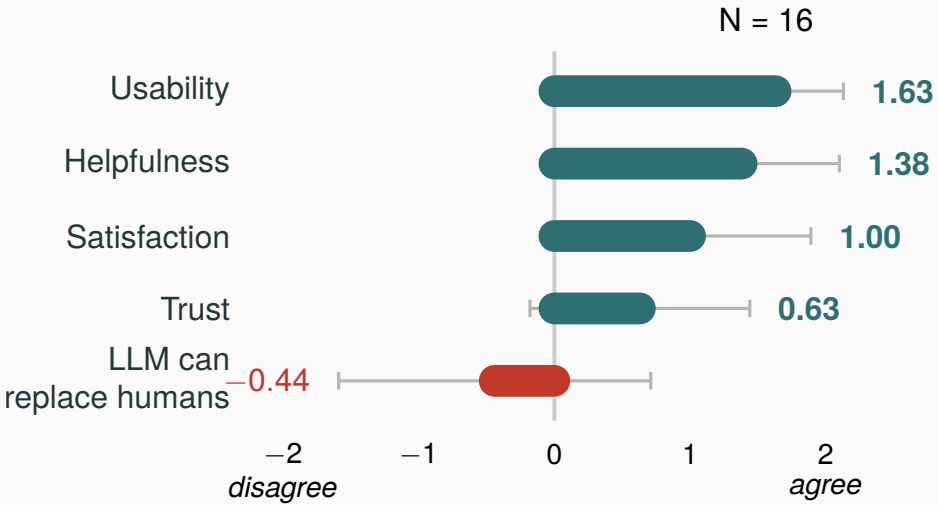

Researchers find UXAgent usable and helpful — but reject it as a *replacement* for humans.

Protocol Iteration Outcomes

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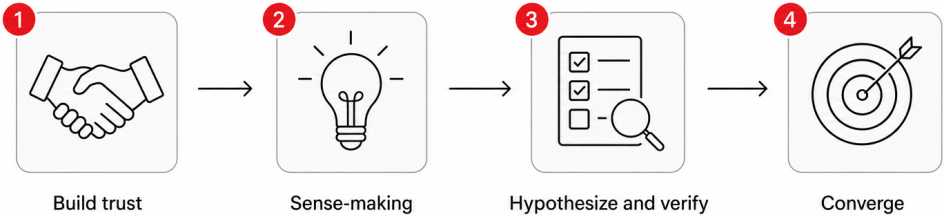
14/16

proposed further study-design changes after using UXAgent

Typical Iteration Cadence



> *I trust it pretty much because it's aligned with the reasoning.*
>
> **— P3**

Researchers cross-check **reasoning** $\leftrightarrow$ **action** $\leftrightarrow$ **screenshot** before trusting the data.
Trust rated $M = 0.62$ (SD 0.81).

Is the behavior human-like? Mixed.

convincing

"The reasoning of the actions is very reasonable."

— P7

not convincing

"I just do not think a real human participant will have actions like that."

— P6

. . . but "users always surprise me" (**P8**) — real users are unpredictable too, so some surprising agent behavior may feel *authentic* rather than less.

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“

I’m really afraid that researchers only rely on this and overtrust such data.

— P1

Most disagreed that agents can **replace** humans ($M = -0.44$).
Position: **augment** human studies — best for early, low-cost iteration.

We’ve seen UXAgent can help UX researchers ...

We've seen UXAgent can help UX researchers ...

Remaining question: How accurate are LLM Agents in simulating human behavior?

Case Study 2: Use Real Online Customer Behavior Data to Evaluate and Improve

Case Study 2: Use Real Online Customer Behavior Data to Evaluate and Improve

Task & Method

- We focus on the **human behavior simulation task**
  - Generate the next user action based on the context and past actions.
  - Specifically, in the online shopping scenario.

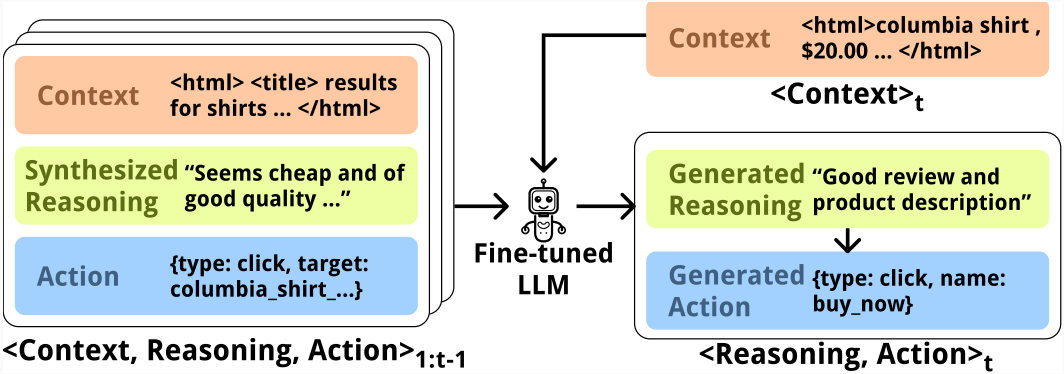

Figure 5: Overview of the next action prediction task.

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230,965

real user actions

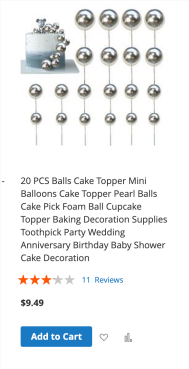
31,865 sessions **3,526** users

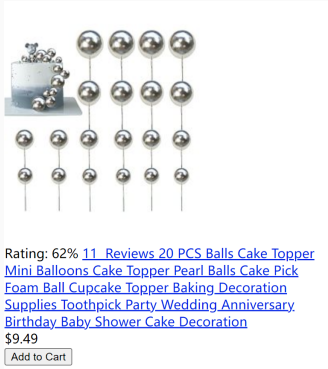
How sessions end

4,432 purchases

27,433 terminations

Collected from a real-world online shopping platform.





Rendered page

Context: simplified HTML

JS/CSS stripped, page structure kept; each interactable element gets a unique name (e.g. `product_form.add_to_cart`).

Action = the next *raw browser action* by the user

`click`

click on an element

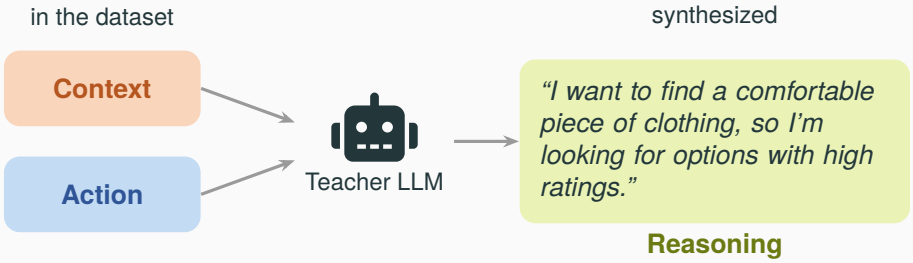
`type_and_submit`

type text and
press Enter

`terminate`

close the
browser window

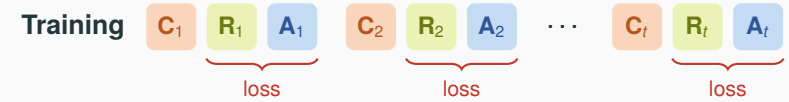
Not shopping-specific: the same setup generalizes to other web domains.



✔ makes each action explainable

✘ missing from existing datasets

Fine-tune the LLM to predict the next ⟨Reasoning, Action⟩



whole session in one pass · loss only on Reasoning and Action tokens

Case Study 2: Use Real Online Customer Behavior Data to Evaluate and Improve

Evaluation and Experiments

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Next Action · Accuracy

predicted	ground truth	
`click·add_to_cart`	`click·add_to_cart`	✓
`click·reviews`	`click·add_to_cart`	✗

action type *and* target must match

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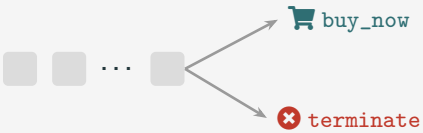
Next Action · Accuracy

predicted	ground truth	
click·add_to_cart	click·add_to_cart	✓
click·reviews	click·add_to_cart	✗

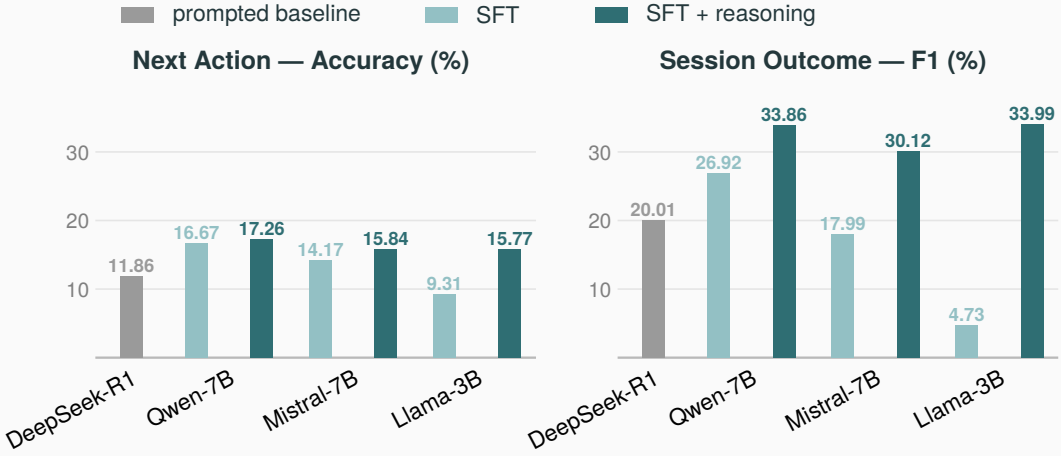
action type *and* target must match

Session Outcome · F1



predict how the session ends

Results – Accuracy & F1



Prompting is not all you need: small fine-tuned models beat the best prompted model, and synthesized *reasoning* lifts outcome F1 further.

Results – Distribution

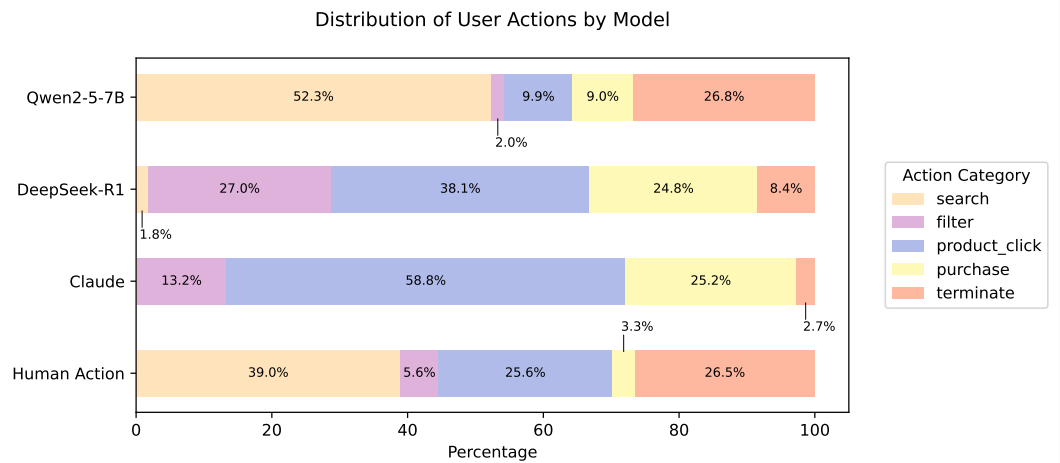


Figure 6: Distribution of the action types in the dataset.

SFT Imitates, RL Explores

Case Study 3: Making Agents RL-Ready: an Environment to Act in, and Verified Rewards to Learn from

RL needs two things:

a place to act and a signal to learn from

Case Study 3: Making Agents RL-Ready: an Environment to Act in, and Verified Rewards to Learn from

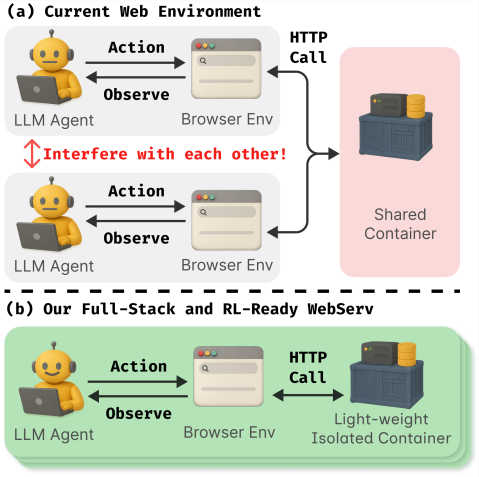
WebServ: A Browser-Server Environment for RL Web Agents

Web Browsers are NOT RL-Ready

Per-site setup

Noisy pages

Web Servers are NOT RL-Ready


(a) Current Web Environment
Action
Observe
LLM Agent
Browser Env
HTTP Call
Interfere with each other!
Action
Observe
LLM Agent
Browser Env
Shared Container
(b) Our Full-Stack and RL-Ready WebServ
Action
Observe
LLM Agent
Browser Env
HTTP Call
Light-weight Isolated Container

Evaluation Highlights

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- WebServ + Claude 4.5: 40.0% (Shopping), 62.9% (CMS), 50.0% (GitLab) on WebArena-Lite.
- Faster spin-up: launch latency $\sim$**5**$\times$ faster.
- Storage: $\sim$**240**$\times$ smaller persistent storage.
- 200+ concurrent containers on a single host.

Benchmarks vs. Vanilla WebArena

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Environment	Model	Shopping	CMS	GitLab
Vanilla WebArena	GPT-4o	11.1	20.0	10.0
Vanilla WebArena	OpenAI-o3	33.3	45.7	46.7
Vanilla WebArena	Llama-3.1-8B	8.9	5.7	10.0
WebServ	GPT-4o	20.0	28.6	43.3
WebServ	OpenAI-o3	37.8	48.6	46.7
WebServ	Llama-3.1-8B	11.1	11.4	16.7

Table 1: Single-prompt success rate (%) on WebArena-Lite. WebServ lifts accuracy across proprietary and open-source models.

Metric	WebServ (Incus)	Naïve Docker
Launch speed	1.781 s	8.963 s
Storage	28.01 MiB	6.78 GiB
Memory	1.74 GiB	1.63 GiB

Table 2: Incus-based containers cut launch latency ($\sim 5\times$) and storage ($\sim 240\times$) at similar RAM.

WebServ gives agents a place to act.

WebServ gives agents a place to act.

Where does a verified learning signal come from?

Case Study 3: Making Agents RL-Ready: an Environment to Act in, and Verified Rewards to Learn from

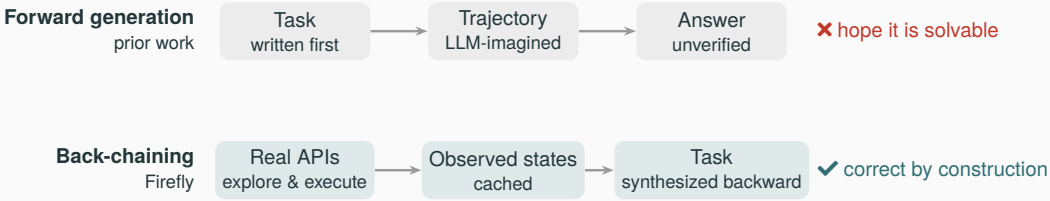
Firefly: Verified Tool-Call Data from Real APIs

Scalable, Real, Verifiable: Pick Two

✗ not scalable ✗ not real ✗ not verifiable

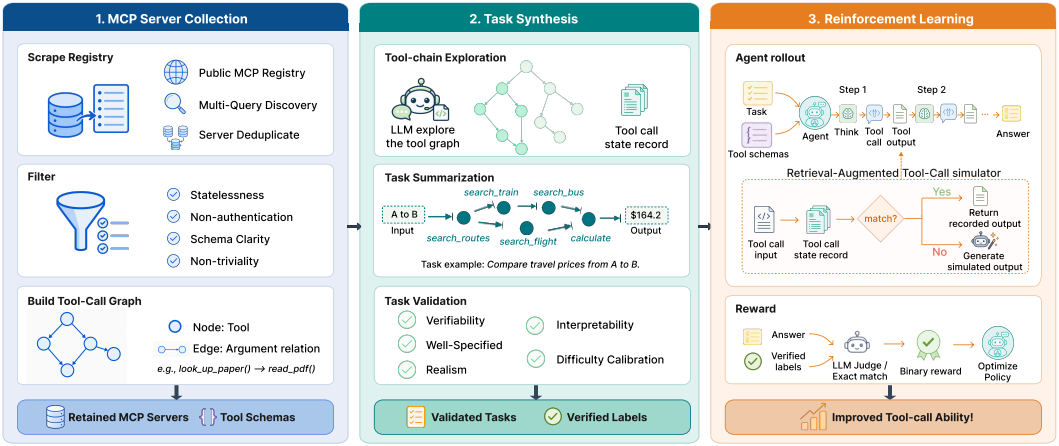
How can we get tool-call training data that is scalable, real, and verifiable at the same time?

Key Idea: Invert the Pipeline



Correctness is built into *generation*, not bolted on by post-hoc filtering.

Firefly Pipeline



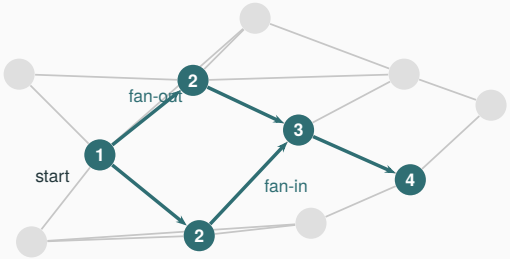
Filter real servers → explore and back-chain tasks → RL in an offline simulator.

Simple Idea, Two Challenges at Scale

Hundreds of servers, ∼1,000 tools: blind exploration wastes most calls.
→ **graph-guided exploration**

Live APIs change, go down, rate-limit: rollouts are not reproducible.
→ **replay simulator**

Graph-Guided Exploration



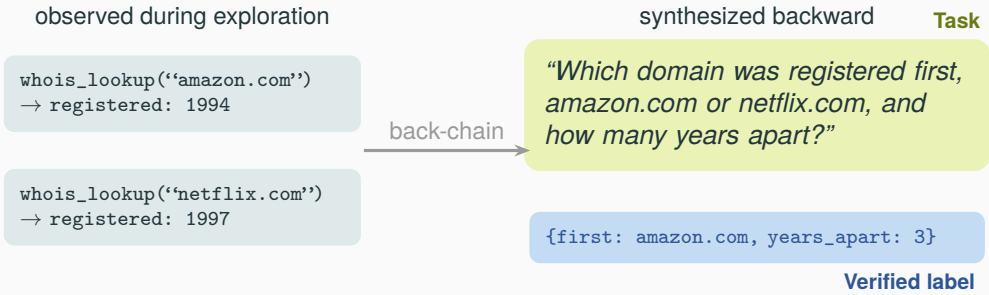
993 tools from **240** real MCP servers

83K edges: "can *A*'s output feed *B*?" judged by an LLM from tool schemas

A strong LLM walks a sampled sub-graph, **executing real tools** at every step, and leaves behind a DAG of actual calls.

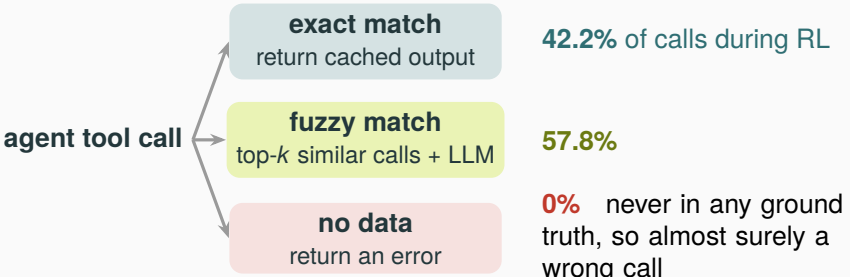
Without the graph, most calls land on incoherent tool combinations.

Back-Chaining: From Observed States to a Task



Every answer field comes from a response that *actually happened*.
Unused nodes become distractors. An LLM judge drops about half the candidates.

Real APIs Drift: Replay Them Instead



Cache every exploration call; replay it instead of hitting live servers during RL.

Same tension as WebServ: real execution's realism, a replay's reproducibility.

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5,144
verified tasks

240 servers **993** tools

3.0 tool calls per task on average (1–10)

38.5% of tasks span *multiple* servers

4.6 answer fields per task

7.8 / 10 realism, LLM-judged

One-time cost: 23.5B tokens, about $47K. After that, offline rollouts are free.

Compared with Existing Datasets

Northeastern University

Dataset	Tasks	Servers/APIs	Multi-step	Verifiable	Real APIs
API-Bank	1,888	2,138	✓	✓	×
ToolAlpaca	3,938	400+	×	×	✓
APIGen-MT	5,000	27	✓	✓	×
Tau2-Bench	279	68	✓	✓	×
Toucan	1.5M	2000+	✓	×	✓
Firefly	**5,144**	**993**	✓	✓	✓

The only one that is multi-step, verifiable, and grounded in real APIs.

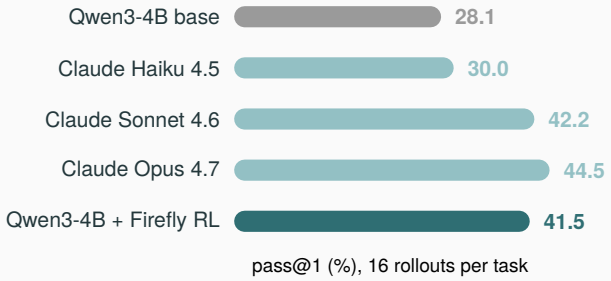
Results: A 4B Model Matches Claude Sonnet

Northeastern University

41.5

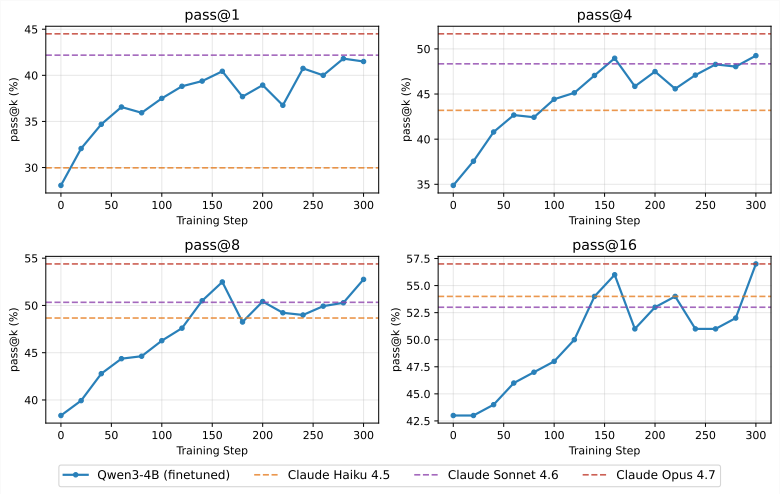
pass@1 on held-out Firefly tasks
up from 28.1 before RL

Qwen3-4B after GRPO on Firefly matches **Claude Sonnet 4.6** (42.2)



At pass@8 it reaches 52.8, above Sonnet (50.3) and close to Opus (54.4).

Results: Steady Gains over RL Training



Dashed lines: Claude Haiku 4.5, Sonnet 4.6, Opus 4.7, same protocol.

Results: It Transfers

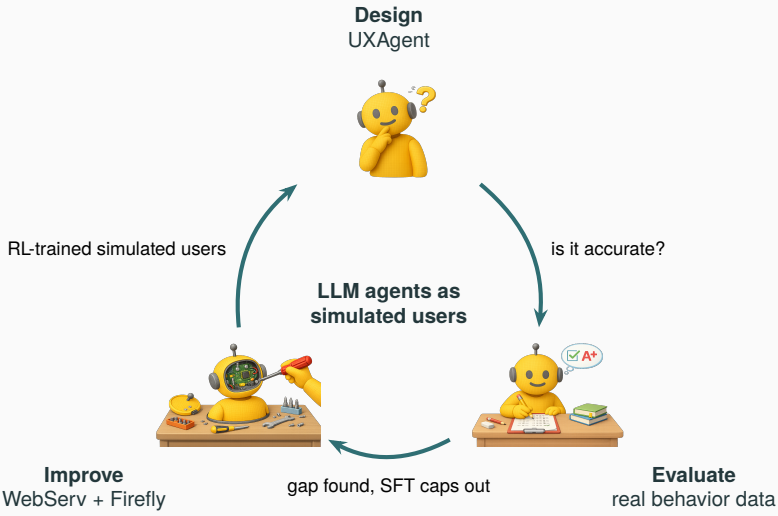
Northeastern University

Model	Tau2-Bench			MCP-Atlas	MCPMark		
	Retail	Airline	Telecom		File Sys.	PG Easy	PG Std
Qwen3-4B	49.1	36.5	18.9	19.4	40.0	70.0	9.5
+ **Firefly RL**	**62.7**	**52.5**	**20.4**	**26.0**	**60.0**	**80.0**	**13.3**

Improves on every public benchmark, **multi-turn** and **real MCP execution** included.

Trained only on single-turn tasks in a replay simulator: not memorization.

Closing the Loop



- Designing: UXAgent lets UX researchers iterate on a study design before spending participants' time.
- Evaluating: against real behavioral data, prompted agents are far from human; fine-tuning with reasoning helps, but SFT can only imitate.
- Improving: WebServ gives agents a scalable place to act, Firefly a verified signal to learn from; RL takes a 4B model to Claude Sonnet level.

arXiv:2608.11977v1 [cs.AI] 12 Aug 2026

Retry, Switch, or Abstain? Learning Strategy-Aware Tool-Use Policies via Controlled Error Injection

arXiv:2606.12908v1 [cs.CL] 11 Jun 2026

SENTINEL: Failure-Driven Reinforcement Learning for Training Tool-Using Language Model Agents

arXiv:2606.03099v2 [cs.CL] 6 Jun 2026

CollabSim: A CSCW-Grounded Methodology for Investigating Collaborative Competence of LLM Agents through Controlled Multi-Agent Experiments

arXiv:2510.07230v2 [cs.CL] 18 Oct 2025

Customer-R1: Personalized Simulation of Human Behaviors via RL-based LLM Agent in Online Shopping

arXiv:2510.19245v1 [cs.CY] 29 Oct 2025

See, Think, Act: Online Shopper Behavior Simulation with VLM Agents

arXiv:2509.21501v1 [cs.HC] 25 Sep 2025

DPRF: A Generalizable Dynamic Persona Refinement Framework for Optimizing Behavior Alignment Between Personalized LLM Role-Playing Agents and Humans

Bingsheng Yao
Northeastern University

Bo Sun
Northeastern University

Yuanzhe Dong
Stanford University

arXiv:2509.03590v2 [cs.HC] 27 Mar 2026

LLM Agent Meets Agentic AI: Can LLM Agents Simulate Customers to Evaluate Agentic-AI-based Shopping Assistants?

Lu Sun
University of California San Diego

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arXiv:2603.11409v1 [cs.HC] 12 Mar 2026

Characterizing LLM-Empowered Personalized Story-Reading and Interaction for Children: Insights from Multi-Stakeholder Perspectives

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Ellena Pan
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Figure 1: The main interface of StoryStars. F1 displays book content with real-time captions and audio narration. F2 shows the concept with its explanation. F3 presents the conversation between the child and the chatbot, extending from the story content to real-world knowledge based on the concept word. F4 provides accessible input options for children.

Abstract

Storytelling and interaction is highly valued by parents in their story-reading activities with children. While AI-empowered story-reading tools have been increasingly used, their abilities to support personalized interaction with children are still limited. Recent advances in large language models (LLMs) show promise in facilitating personalized interactions, but little is known about how to effectively and

*Work done when visiting Northeastern University.
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arXiv:2509.03590v2 [cs.HC] 27 Mar 2026

Shop-R1: Rewarding LLMs to Simulate Human Behavior in Online Shopping via Reinforcement Learning

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Kun Qian[3] Yuxin Tang[3] Ran Xue[3] Houyu Zhang[3] Qingjun Cui[3]
Yufan Guo[3] Dakuo Wang[4]
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[4]University of Illinois Urbana-Champaign [5]Northwestern University

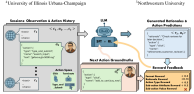

Figure 1: Overview of the proposed reinforcement learning framework, *Shop-R1*, designed to simulate real human behaviors in web-based shopping environments. Given an action history $a_{t-k:t-1}$ with corresponding website observations $o_{t-k:t-1}$, the model predicts the next action a_t and its rationale r_t based on the history and the latest website observation o_t. The generated responses are evaluated by a reward model assessing the correctness and coherence of the rationale, action type, accuracy, and sub-action (attribute and value) accuracy.

ABSTRACT

Large Language Models (LLMs) have recently demonstrated strong potential in generating 'believable human-like' behavior in web environments. Prior work has explored augmenting training data with LLM-synthesized rationales and applying supervised fine-tuning (SFT) to enhance reasoning ability, which can in turn improve downstream action prediction. However, the performance of such approaches remains inherently bounded by the reasoning capabilities of the model used to generate the rationales. In this paper, we introduce *Shop-R1*, a novel reinforcement learning (RL) framework aimed at enhancing the reasoning ability of LLMs for simulation of real human behavior in online shopping environments. Specifically, *Shop-R1* decomposes the human behavior simulation task into two stages: rationale generation and action prediction, each guided by distinct reward signals. For rationale generation, we leverage self-supervised signals derived from distributional to guide the reasoning process in a self-supervised manner. For

arXiv:2507.17834v2 [cs.CL] 23 Feb 2026

PADFormer: Pose-agnostic Anomaly Detection from Sparse View Images

arXiv:2606.01506v7 [cs.CL] 17 May 2026

Scale Up Strategically: Learning Compositional Generalization via Bias-Aware Evaluation and Data Collection for Robotic Manipulation

arXiv:2507.21028v1 [cs.CL] 28 Jul 2025

Humans' Almanac: A Human Collaboration Dataset of Action-Level Mental Model Annotations for Agent Collaboration

Jiajie Chen
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Yuxuan Lu
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arXiv:2601.20154v3 [cs.CL] 21 Apr 2026

SpanUQ: Span-Level Uncertainty Quantification for Large Language Model Generation

arXiv:2604.19571v2 [cs.CL] 28 Nov 2026

Trajectory2Task: Training Robust Tool-Calling Agents with Synthesized Yet Verifiable Data for Complex User Intents

arXiv:2607.21582v1 [cs.RO] 23 Jul 2026

Multi-Agent-as-Judge: Aligning LLM-Agent-Based Automated Evaluation with Multi-Dimensional Human Evaluation

Jiaju Chen
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Xiaojie Wang
Amazon

arXiv:2606.06383v2 [cs.AI] 6 Jun 2026

OPeRA: A Dataset of Observation, Persona, Rationale, and Action for Evaluating LLMs on Human Online Shopping Behavior Simulation

Ziyi Wang*, Yuxuan Lu*, Wenbo Li†, Amirali Amini†,
Bo Sun†, Yakov Bart†, Weimin Lyu†, Jiri Gesi†,
Tian Wang†, Jing Huang†, Yu Su†, Upol Ehsan†

Thank you!
Questions?